

Program: Diploma in Fire Technology and Safety	
Course Code: 6461D	Course Title: Advanced Fire Control Technology
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the fundamental principles of fire chemistry and fire dynamics.
- To learn the theories and mechanisms of fire behavior, including pre- and post-flashover fires.
- To apply fire modeling techniques in real-world scenarios, particularly in warehouse and enclosure environments.
- To explore combustion toxicity, explosion phenomena, and their effects on fire control strategies.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4
Safety in Engineering Industries		Safety in Engineering Industries	4

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	CognitiveLevel
CO1	Understand fire chemistry, dynamics, and the principles of fire behavior and spread.	15	Understanding
CO2	Analyze fire dynamics and phenomena, including plume theory and flashover mechanisms.	14	Applying
CO3	Implement fire modeling techniques and understand their applications and limitations.	14	Analyzing
CO4	Assess explosion phenomena and evaluate combustion toxicity in fire scenarios.	15	Evaluating
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3				
CO2	3	3	3				
CO3	3	3	2				
CO4	3	2	3	3			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand fire chemistry, dynamics, and the principles of fire behavior and spread.		
M1.01	Fire chemistry: Fire tetrahedron, stoichiometry, heat of combustion, ignition.	4	Understanding

M1.02	Flammability characteristics of fuels, ignition limits.	3	Understanding
M1.03	Plume theory: Physical dimensions, entrainment, and hot gas layer descent.	4	Applying
M1.04	Pre-flashover fire dynamics: Ignition, flame spread theories, plume-driven and ventilation-controlled fires.	4	Applying
CO2	Analyze fire dynamics and phenomena, including plume theory and flashover mechanisms.		
M2.01	Post-flashover compartment fires: Flashover theories, fully developed fires.	3	Understanding
M2.02	Decay period and traveling fires: Dynamics and characteristics.	4	Applying
M2.03	Fire growth and transition from pre-flashover to post-flashover.	4	Analyzing
M2.04	Compartment fire behavior: Ventilation-controlled and plume-driven fires.	3	Applying
	SeriesTestI	1	
Contents: Flashover and Fire Development: Understanding the transition from ignition to flashover and beyond. Fully Developed Fires: Characteristics of fires after flashover, including decay and traveling fires. Ventilation-Controlled Fires: How ventilation affects the behavior of compartment fires.			
CO3	Implement fire modeling techniques and understand their applications and limitations.		
M3.01	Introduction to premixed and diffusion flames.	3	Understanding
M3.02	Fire modeling: Zone modeling (CFAST), field modeling (FDS), and their applications	4	Applying
M3.03	Empirical correlations: Flame height, heat release rate, species yields.	4	Analyzing
M3.04	Applications of fire modeling in warehouse storages.	3	Applying

Contents:

Flame Behavior: Understanding premixed and diffusion flames.
Fire Modeling Techniques: Introduction to zone modeling (CFAST) and field modeling (FDS).
Empirical Correlations: Correlating fire-related data like flame height, heat release, and species yields.
Applications: Real-world applications of fire models, particularly in warehouse storage systems.

CO4	Assess explosion phenomena and evaluate combustion toxicity in fire scenarios.		
M4.01	Explosion phenomena: Backdrafts and smoke explosions.	3	Understanding
M4.02	Factors affecting the severity of explosions.	4	Analyzing
M4.03	Combustion toxicity: Major products, exposure limits, and evaluation methods.	4	Applying
M4.04	Evaluating combustion toxicity and risk in fire scenarios.	4	Evaluating
	SeriesTest II	1	

Contents:

Explosion Phenomena: Understanding the occurrence and severity of backdrafts and smoke explosions.
Factors Affecting Explosion Severity: Key parameters that influence the intensity of explosions.
Combustion Toxicity: Identifying harmful combustion products, their exposure limits, and evaluation methods.

Text/Reference:

T/R	BookTitle/Author
R1	SFPE Handbook of Fire Protection Engineering, 5th Edition, NFPA, 2016.
R2	Drysdale, D., An Introduction to Fire Dynamics, 3rd Edition, Wiley, 2011
R3	Sikanen and Hostikka, Fire Dynamics and Modelling – Small Enclosures, Wiley, 2020
R4	Kuligowski et al., The SFPE Guide to Human Behavior in Fire, SFPE, 2011

R5	Bianchini, G., Fire Risk Management to Prevent Disaster, Springer, 2020.
R6	Barrow, R., Combustion Toxicology, Taylor & Francis, 1992.